Light

GCSE Computer Science — Programming Data Structures

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is an array (or list) in programming?

- ○ **A.** A variable that stores a single value
- ○ **B.** A function that returns multiple results
- ○ **C.** A loop that processes multiple items
- ○ **D.** A data structure that stores multiple values under a single identifier

Q2 (1 mark)

In most programming languages, what is the index of the first element in an array?

- A. 0
- B. 1
- C. -1
- D. The array name

Q3 (1 mark)

Consider the following code:

```
word = "Python"
print(word[0])
```

What is output by this code?

- A. Python
- B. P
- C. y
- D. 0

Q4 (1 mark)

What does the `len()` function return when applied to a string?

- A. The last character of the string
- B. The index of the last character
- C. The number of characters in the string

- ○ **D.** The string converted to lowercase

Q5 (1 mark)

Which method is used to add an element to the end of a list in Python?

- ○ **A.** add()
- ○ **B.** append()
- ○ **C.** insert()
- ○ **D.** extend()

Q6 (1 mark)

Consider the following code:

```
text = "hello"
result = text.upper()
print(result)
```

What is output by this code?

- ○ **A.** hello
- ○ **B.** HELLO
- ○ **C.** Hello
- ○ **D.** hELLO

Q7 (1 mark)

What does the `lower()` method do when applied to a string?

- ○ **A.** Removes all lowercase letters
- ○ **B.** Changes the original string to lowercase
- ○ **C.** Returns the lowest character by ASCII value
- ○ **D.** Returns a new string with all characters in lowercase

Q8 (1 mark)

Consider the following code:

```
items = ["a", "b", "c", "d"]
print(items[-1])
```

What is output by this code?

- ○ **A.** a
- ○ **B.** b
- ○ **C.** d
- ○ **D.** Error

Q9 (1 mark)

What is a record (dictionary) in programming?

- ○ **A.** A list that can only store numbers

- ○ **B.** A variable that holds a single text value
- ○ **C.** A loop that processes data repeatedly
- ○ **D.** A data structure that stores key-value pairs

Q10 (1 mark)

Which data type is most appropriate for storing a product price like £19.99?

- ○ **A.** Integer
- ○ **B.** String
- ○ **C.** Boolean
- ○ **D.** Real (float)

Q11 (1 mark)

What is the character (char) data type used to store?

- ○ **A.** A sequence of multiple characters
- ○ **B.** A whole number
- ○ **C.** A True or False value
- ○ **D.** A single letter, digit, or symbol

Q12 (1 mark)

Consider the following code:

```
numbers = [10, 20, 30]
for n in numbers:
    print(n)
```

How many times does the loop execute?

- A. 1
- B. 2
- C. 3
- D. 30

Q13 (2 marks)

Consider the following code:

```
text = "Computer"
print(text[0:4])
```

What is output by this code?

- A. Comp
- B. Compu
- C. ompu
- D. Computer

Q14 (2 marks)

Consider the following code:

```
word = "Programming"
print(word[4:])
```

What is output by this code?

- ○ **A.** Prog
- ○ **B.** ramming
- ○ **C.** gramming
- ○ **D.** amming

Q15 (2 marks)

Consider the following code:

```
colours = ["red", "green", "blue"]
colours[1] = "yellow"
print(colours)
```

What is output by this code?

- ○ **A.** ["red", "green", "blue"]
- ○ **B.** ["yellow", "green", "blue"]
- ○ **C.** ["red", "yellow", "blue"]
- ○ **D.** ["red", "green", "yellow"]

Q16 (2 marks)

Consider the following code:

```
student = {"name": "Emma", "age": 15, "grade": "B"}
print(student["age"])
```

What is output by this code?

- A. age
- B. "age": 15
- C. Emma
- D. 15

Q17 (2 marks)

Consider the following code:

```
grid = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]
print(grid[1][2])
```

What is output by this code?

- A. 2
- B. 4
- C. 6
- D. 8

Q18 (2 marks)

Consider the following code:

```
fruits = ["apple", "banana", "cherry", "banana"]
fruits.remove("banana")
```

```
print(fruits)
```

What is output by this code?

- A. ["apple", "cherry"]
- B. ["apple", "banana", "cherry"]
- C. Error
- D. ["apple", "cherry", "banana"]

Q19 (2 marks)

Consider the following code:

```
message = "Hello World"
position = message.find("o")
print(position)
```

What is output by this code?

- A. 2
- B. 4
- C. 7
- D. -1

Q20 (2 marks)

Consider the following code:

```
scores = [85, 92, 78, 95, 88]
print(len(scores))
```

What is output by this code?

- A. 5
- B. 438
- C. 95
- D. 4

Q21 (2 marks)

Consider the following code:

```
values = [3, 7, 2, 5]
total = 0
for v in values:
    total = total + v
print(total)
```

What is output by this code?

- A. 17
- B. 3725
- C. 4
- D. 0

Q22 (2 marks)

A CSV file contains the line:

```
Smith,John,25,London
```

After reading this line into a variable called `line`, which code correctly splits it into a list of fields?

- ○ **A.** fields = line.split(",")
- ○ **B.** fields = line.divide(",")
- ○ **C.** fields = split(line, ",")
- ○ **D.** fields = line.separate(",")

Q23 (2 marks)

Consider the following code:

```
name = "Alice"
age = 16
message = name + " is " + str(age) + " years old"
print(message)
```

What is output by this code?

- ○ **A.** Alice is 16 years old
- ○ **B.** name is age years old
- ○ **C.** Error: cannot concatenate
- ○ **D.** Alice is str(age) years old

Q24 (3 marks)

Consider the following code:

```
table = [[1, 2], [3, 4], [5, 6]]
total = 0
for row in table:
```

```
    for item in row:
        total = total + item
print(total)
```

What is output by this code?

- A. 6
- B. 15
- C. 21
- D. 12

Q25 (3 marks)

Consider the following code:

```
numbers = [34, 67, 23, 89, 45]
largest = numbers[0]
for n in numbers:
    if n > largest:
        largest = n
print(largest)
```

What is output by this code?

- A. 34
- B. 45
- C. 89
- D. 258

Q26 (3 marks)

Consider the following code:

```
book = {"title": "1984", "author": "Orwell", "year": 1949}
book["pages"] = 328
book["year"] = 1948
print(book)
```

What is output by this code?

- ○ **A.** {"title": "1984", "author": "Orwell", "year": 1949}
- ○ **B.** {"title": "1984", "author": "Orwell", "year": 1948, "pages": 328}
- ○ **C.** {"title": "1984", "author": "Orwell", "year": 1949, "pages": 328}
- ○ **D.** Error: cannot add new keys

Q27 (3 marks)

Consider the following code:

```
text = "Programming"
first = text[:3]
middle = text[3:7]
last = text[7:]
result = last + middle + first
print(result)
```

What is output by this code?

- ○ **A.** minggrammPro
- ○ **B.** PrograProgramming
- ○ **C.** ProgrammingProgramming
- ○ **D.** minggramPro

Q28 (3 marks)

Consider the following code:

```
numbers = [12, 7, 19, 4, 23, 8]
result = []
for n in numbers:
    if n > 10:
        result.append(n)
print(result)
```

What is output by this code?

- A. [12, 19, 23]
- B. [7, 4, 8]
- C. [12, 7, 19, 4, 23, 8]
- D. 3

Q29 (3 marks)

Consider the following code:

```
matrix = [[0, 0, 0], [0, 0, 0], [0, 0, 0]]
for i in range(3):
    matrix[i][i] = 1
print(matrix)
```

What is output by this code?

- A. [[1, 0, 0], [0, 1, 0], [0, 0, 1]]
- B. [[1, 1, 1], [1, 1, 1], [1, 1, 1]]
- C. [[1, 0, 0], [1, 0, 0], [1, 0, 0]]

- D. [[0, 0, 1], [0, 1, 0], [1, 0, 0]]

Q30 (3 marks)

Consider the following code:

```
students = [
    {"name": "Ali", "score": 72},
    {"name": "Beth", "score": 85},
    {"name": "Carl", "score": 68}
]
total = 0
for student in students:
    total = total + student["score"]
average = total / len(students)
print(average)
```

What is output by this code?

- A. 75
- B. 75.0
- C. 225
- D. 3

Total: **55** marks • Time: **30 min**

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